This assignment will not be graded and is only for practice.

1. Key points:

Rank of a matrix:
- Maximum number of linearly independent column vectors (or row vectors) in a matrix.

NOTE: Rank of a matrix is always less than or equal to the number of columns (or rows).

How to find rank of a matrix A:
- Reduce the matrix to row echelon form.
- Find the number of non zero rows in the reduced matrix which will be the rank of the matrix A.

1) Consider the following two matrices: ***1 point***

P = $\begin{bmatrix} 1 & 0 & 3 \\ 2 & -2 & 0 \\ 0 & 1 & -1 \\ 0 & 2 & 1 \end{bmatrix}$; Q = $\begin{bmatrix} 1 & 1 & -2 & 0 \\ 1 & 0 & 2 & 0 \\ 0 & -1 & 0 & 4 \end{bmatrix}$

Which of the following statements is (are) TRUE? [Hint: Try to reduce the matrices to row echelon form and find the number of non zero rows]

- [ ] Rank(P) = 4
- [ ] Rank(P) = Rank(Q)
- [ ] Rank(QP) = 4
- [ ] Rank(PQ) = Rank(QP)

No, the answer is incorrect.
Score: 0

Accepted Answers:

Rank(P) = Rank(Q)
Rank(PQ) = Rank(QP)

2 Key points:

Null space of a matrix A:

- Let A be an $m \times n$ matrix. The subspace W = $\{x \mid x \in \mathrm{R}^n, Ax = 0\}$ is called the null space of A.
- The dimension of the null space W is called the nullity of A.

Rank-Nullity theorem:

- Let A be an $m \times n$ matrix, then rank(A) + nullity(A) = n.

Consider the following matrix:

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Use the matrix A to answer the following questions (2) and (3).

2) Which of the following vector space represents the null space of A? ***1 point***

- $\{(x_1, 0, x_2) | x_1, x_2 \in \mathrm{R}\}$
- $\{(0, x_1, 0) | x_1 \in \mathrm{R}\}$
- $\{(x_1, x_2, 0) | x_1, x_2 \in R\}$
- $\{(0, 0, x_1) | x_1 \in \mathrm{R}\}$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\{(x_1, 0, x_2) | x_1, x_2 \in \mathrm{R}\}$

3) What will be the rank of the matrix A?

[Hint: Find the nullity of A and use Rank-Nullity theorem]

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 1

1 point

3 Key points:

Linear transformation:
A function f from one vector space V to another vector space W $(f : V \longrightarrow W)$ is said to be a linear transformation if for any two vectors v_1 and v_2 in the vector space V and for any $c \in \mathbb{R}$ (scalar) the following conditions hold:
(i) $f(v_1 + v_2) = f(v_1) + f(v_2)$
(ii) $f(cv_1) = cf(v_1)$

4) In a super market, the rates of a pen, a water bottle, a tooth brush, and a chocolate are ₹ 10, ₹ 40, ₹ 25, and ₹ 20 respectively. Suppose $T(x, y, z, w)$ denotes the total cost of x pens, y water bottles, z tooth brushes, and w chocolates. Which of the following will be the correct expression for $T(x, y, z, w)$? ***1 point***

- ○ $T(x, y, z, w) = x + y + z + w$
- ○ $T(x, y, z, w) = 20x + 25y + 40z + 10w$
- ○ $T(x, y, z, w) = (x + y + z + w)(10 + 40 + 25 + 20)$
- ○ $T(x, y, z, w) = 10x + 40y + 25z + 20w$

No, the answer is incorrect.
Score: 0

Feedback:

Check whether by evaluating for particular values of x, y, z, w the function value matches the cost.

Accepted Answers:

$T(x, y, z, w) = 10x + 40y + 25z + 20w$

5) Suppose Keerthana bought 3 pens, 5 water bottles, 10 chocolates and she did not buy any tooth brushes. Which of the following options denotes the total cost paid by her? ***1 point***

- ☐ $T(3, 5, 1, 10)$
- ☐ $T(3, 5, 0, 10)$

- [ ] ₹430
- [ ] ₹450

No, the answer is incorrect.

Score: 0

Accepted Answers:

$T(3, 5, 0, 10)$
₹430

Your score is: 0/5